

OSAZEE ERO

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PROFESSIONAL SUMMARY

AI & Machine Learning Engineer with industry and doctoral research experience spanning generative AI, agentic workflows, engineering simulation, and computer vision. Develops RAG pipelines and LLM-assisted services using Python, FastAPI, Docker, and AWS. Built and deployed Contextly, a document intelligence application with hybrid retrieval, source citations, authenticated access, and evaluation workflows. PhD in Mechanical and Mechatronics Engineering from the University of Waterloo.

TECHNICAL SKILLS

- **Software & data:** Python, SQL, TypeScript, pandas, NumPy, PostgreSQL, Git, REST APIs.
- **Machine learning & deep learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, U-Net, ConvLSTM; classification, regression, image segmentation, anomaly detection, model evaluation.
- **Generative AI & retrieval:** LangChain, LlamaIndex, OpenAI API, Pinecone, pgvector; RAG, embeddings, hybrid search, prompt engineering, tool/function calling, LLM evaluation; LoRA/QLoRA and Hugging Face.
- **Agentic AI:** OpenAI Agents SDK, LangGraph, CrewAI, AutoGen, Model Context Protocol (MCP), multi-agent orchestration.
- **MLOps & application delivery:** FastAPI, Docker, AWS ECS Fargate, S3, RDS, MLflow, Next.js, Vercel, automated testing, CI/CD, logging.
- **Cloud & observability:** AWS Bedrock, SageMaker, Lambda, API Gateway, SQS, Terraform, GitHub Actions, Langfuse, guardrails; Azure and GCP deployment workflows.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer / Researcher

Oct 2024 - Present

PhaseShift Technologies

- Build and evaluate classification and regression models using engineering and simulation data to support materials prediction and accelerate simulation workflows.
- Develop LLM-assisted workflows for simulation code generation and parameter configuration; contribute to agentic workflows connecting ML predictors, physics-based simulation tools, and optimization steps for materials exploration.
- Built a RAG pipeline using LangChain and Pinecone for internal knowledge retrieval.
- Deployed FastAPI-based AI services using Docker and AWS ECS, with evaluation, logging, and validation practices to support reliable delivery.

Machine Learning Research Associate

Sep 2024 - Oct 2025

Multi-Scale Additive Manufacturing Lab, University of Waterloo

- Developed computer vision and spatiotemporal deep learning models for manufacturing process monitoring and anomaly detection using image, video, and sensor data.
- Automated image and video data preparation workflows; applied quantization and pruning to optimize models for low-latency inference.

Lead Software Developer

Jan 2023 - Jun 2024

OptiFab

- Developed modular Python-based 3D simulation tools for laser scan-path optimization, translating manufacturing requirements into software for comparing scanning strategies.
- Integrated ML components into engineering software, containerized applications with Docker, and introduced unit testing and CI/CD practices.

PhD Researcher - Machine Learning & Computer Vision

Sep 2020 - Jul 2024

University of Waterloo

- Developed self-organizing map and U-Net segmentation approaches to identify defect-related patterns in optical tomography images from laser powder bed fusion.
- Combined fuzzy logic and machine learning for porosity detection, validated predictions against computed tomography measurements, and published two peer-reviewed journal papers.

SELECTED PROJECTS

Contextly | Deployed Document Intelligence Application

Live: contextly.osazeeero.com | Code: github.com/osazee-ero/contextly

- Built an end-to-end PDF question-answering application using Next.js, FastAPI, PostgreSQL/pgvector, and OpenAI, with page-aware ingestion and document/page citations.
- Combined semantic vector search and PostgreSQL full-text search using Reciprocal Rank Fusion; added retrieval and citation evaluation with a 30-question, project-specific dataset.
- Implemented Clerk authentication, account-scoped document access, persistent conversations, usage limits, upload retries, and transaction rollback with quota refunds on failed requests.
- Deployed the frontend on Vercel and the Dockerized API on AWS ECS Fargate, with RDS and private S3 storage; added structured logs and backend/browser regression tests for isolation and failure recovery.

Hand-Controlled Virtual Lab | Computer Vision Prototype

Code: github.com/osazee-ero/AI-powered-Virtual-Lab

- Co-developed a hand-control prototype using an OAK-D camera, TensorFlow, Python, and Unity; connected camera detections to a virtual laboratory over TCP.
- The documented workflow includes hand-detector training, model conversion for OAK-D inference, and basic hand-position interaction in the Unity scene.

EDUCATION

PhD, Mechanical and Mechatronics Engineering | University of Waterloo | 2024

MSc, Systems Engineering | University of Lagos | 2016

BEng, Electrical/Electronic Engineering | University of Benin | 2011

PROFESSIONAL DEVELOPMENT

Completed AI Engineering Training | Udemy | Ed Donner / Ligency

- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents - LLM applications, multimodal systems, Hugging Face, LoRA/QLoRA fine-tuning, and model selection.
- AI Engineer Agentic Track: The Complete Agent & MCP Course - OpenAI Agents SDK, CrewAI, LangGraph, AutoGen, Model Context Protocol (MCP), and multi-agent orchestration.
- AI Engineer Production Track: Deploy LLMs & Agents at Scale - AWS Bedrock, SageMaker, Lambda, SQS, Terraform, GitHub Actions, Langfuse observability, guardrails, and deployments across AWS, Azure, and GCP.

Machine Learning Engineer Nanodegree | Udacity | Completed, 2021

ADDITIONAL EXPERIENCE

Teaching Assistant | University of Waterloo | May 2021 - Oct 2024

- Supported MTE 262 and MTE 325 labs covering digital logic, PLC programming, microprocessor systems, and hardware interfacing.

University Lecturer | University of Benin | Jul 2013 - Aug 2020

- Taught programming and electrical/electronic engineering courses; designed practical assignments and supervised undergraduate engineering projects.

SELECTED PUBLICATIONS

Optical tomography and machine learning for in-situ defects detection in laser powder bed fusion: A self-organizing map and U-Net based approach. Additive Manufacturing, 2023.

An integrated fuzzy logic and machine learning platform for porosity detection using optical tomography imaging during laser powder bed fusion. International Journal of Extreme Manufacturing, 2024.